

42.510 Statistical Learning for Data Science

Model assessment and selection II

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Week 5, Lecture 2

Outline

- Model assessment and model selection
- Bias-variance trade-off
- Subset selection
- Cross validation
- LASSO

Cross-Validation

- Model assessment technique.
- A model is considered good if it has a low *test set error* (*TEST MSE*) .
- We often do not have a large test set to validate our model.
- One method of model assessment here is *Cross Validation*.

Validation set approach

- Divide the data randomly into 2 subsets (often roughly of equal size): the *training set* and the *validation set* or *hold-out set*.
- Use the training set to fit the model, and the validation set to predict and then estimate Mean squared error (MSE).

Potential drawbacks

- The method depends on the points chosen, hence different choices may lead to starkly different estimated MSEs.
- Since we are only using a subset of the available data set, the performance of the model is worse than it would be on a larger data set. And the error estimates tend to be larger.

LOOCV: Leave one out cross validation

Compensates for the drawbacks of the *Validation set approach* yet keeping the same spirit.

- For every $i \in I = \{1, \dots, n\}$, train the model on the set $I \setminus \{i\}$.
- Use this model to predict the i th response, say it is $\hat{y}_i$; and then compute $\text{MSE}_i = (y_i - \hat{y}_i)^2$.
- Compute cross validation error

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i.$$

Advantages

- Less biased, since we are fitting the model to $n - 1$ of the points.
- Doesn't change depending on the random sample, like validation set method.

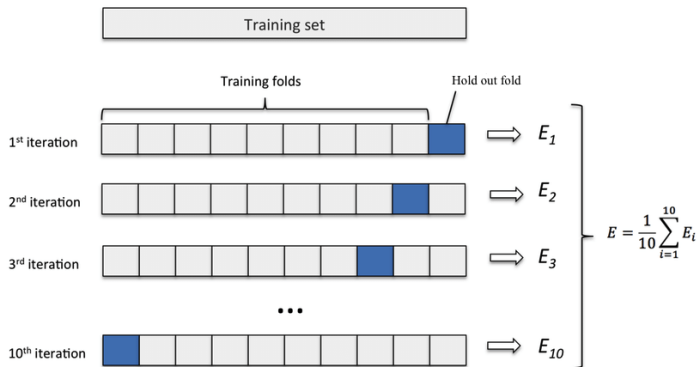
Potential drawback

Computationally intensive: we need to fit n models.

k -fold cross validation

- Divide the data randomly into k subsets (folds) of (roughly) equal size.
- Start with the first fold as a validation set and use the remaining $k - 1$ folds to fit the model.
- Compute the error of the fitted model in the held-out fold.
- Repeat steps 2 and 3 by using the second, third and so on folds as the hold-out fold with the remaining $k - 1$ folds to fit the model.
- Average the error across all the k fitted models to estimate the cross-validation error.
- Some of the commonly used choices for k are 5 or 10.
- When $k = n$, this reduces to LOOCV.

k-fold cross validation



Least Squares (LS)

- Minimize **sum of squared errors (SSE)** in the training set.

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2.$$

- Penalize for complexity of the model. Minimize $\sum_{i=0}^p |\beta_i|$.

LASSO

- LASSO: Least absolute shrinkage and selection operator.
- For a tuning parameter $\lambda \geq 0$:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

- Balance data fit (first term) with model complexity (second term).
- When $\lambda = 0$, LASSO reduces to standard linear regression.
- When $\lambda \uparrow \infty$, the second term dominates and LASSO will make all the beta coefficients for the predictor variables go to zero.

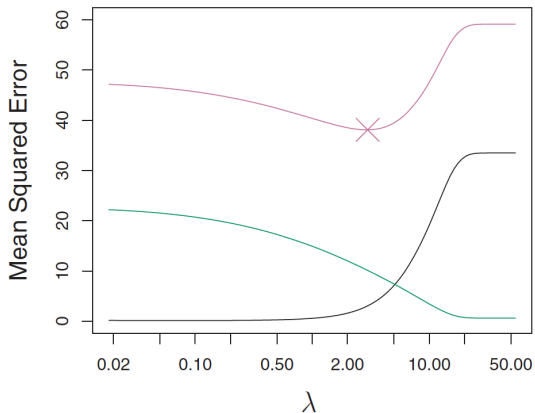
LASSO

- Proposed in the paper *Regression Shrinkage and Selection via the Lasso*, JRSS B, 1996, by *Robert Tibshirani*.
 - ▶ Around 61000 citations as of October 2024.
- Objective function in LASSO is convex. It tries to roughly promote sparsity.
- **Advantage**
 - ▶ Convex objective function $\Rightarrow$ local optimum is the global optimum.
- **Disadvantage**
 - ▶ Objective function is not differentiable, hence no explicit solution.
 - ▶ Still there are efficient ways to solve the problem to optimality.

Choice of λ

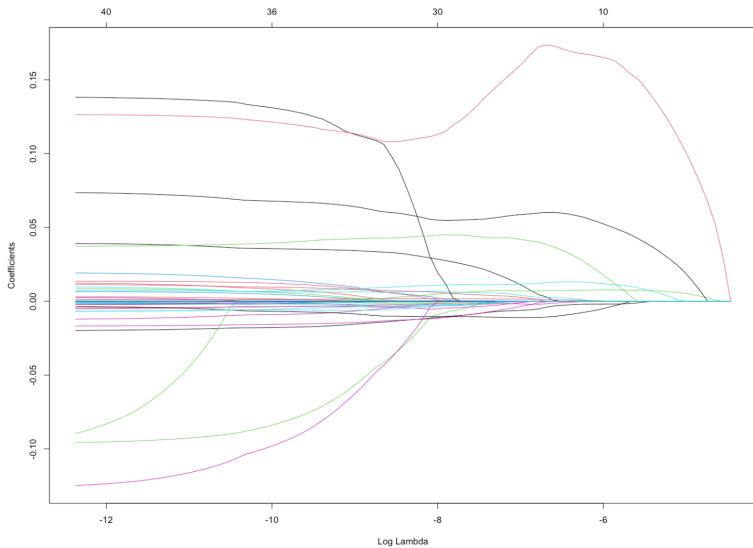
- Use a grid of possible values and compute the CV error for each value of λ .
- Choose the λ with the smallest CV error.
- Finally refit the final model using all observations for the selected value of λ .

LASSO



- Black line: Squared Bias.
- Green line: Variance.
- Purple line: Test MSE.

LASSO



Alternatives to LASSO

- **LASSO**

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

- **Ridge Regression**

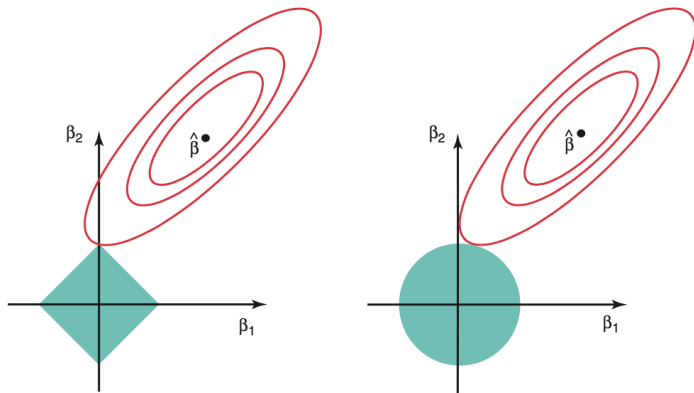
$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 + \lambda \sum_{j=1}^p \beta_j^2.$$

- **Elastic Net**

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2.$$

- ▶ combines ridge regression and LASSO penalty.

LASSO vs Ridge regression



Econometrics: Cross-country growth regression

- Understand factors (economic, political, social) that affect rate of economic growth.
- For example: GDP, degree of capitalism, population growth, equipment investment.
- Robert Barro (1991): Growth rate $\uparrow$
 - ▶ School enrollment rate $\uparrow$
 - ▶ Real per capita GDP (1960 level) $\downarrow$
- Many such variables have been proposed. Little guidance from economic theory on choice.
- Why not use subset selection from linear regression.
- We use dataset from *I just ran two million regressions* by Sala-I-Martin and *Model uncertainty in cross country growth regression* by Fernandez et. al.
- 41 possible explanatory variables with 72 countries.
- Note that if you try all 2^{41} possible combinations, it leads to ~ 2 trillion possibilities.

References

- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 3,7.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 5,6.
- Fernandez et al. (2001), *Model uncertainty in cross country growth regression*.
- Martin (1997), *I just ran two million regressions*.